

4b. General Procedure for Amide Coupling in Batch



General Procedure A

To a 1-dram vial equipped with a PTFE stir bar was added carboxylic acid (1 equiv, 0.5 mmol), *N,N*-dimethylpyridin-4-amine (DMAP) (6.1 mg, 0.1 equiv, 0.05 mmol), and DPDTC (130. mg, 1.05 equiv, 0.525 mmol). 2-MeTHF (1.0 mL, 0.5 M) was added, and the vial was capped and sealed with Teflon tape. The reaction vial was placed into an aluminum heating block which was pre-heated to 95°C and the reaction was stirred vigorously for 30 min. The reaction was taken off the aluminum block, allowed to cool to rt briefly. Once cooled, the cap was removed and amine (1.05 equiv, 0.525 mmol) and 2-MeTHF (0.25 mL, 0.4 M) were quickly added. The vial was capped, taped, and set to stir for another 30 min before working up.

General Procedure B (water soluble amines)

To a 1-dram vial equipped with a PTFE stir bar was added carboxylic acid (1 equiv, 0.5 mmol), DMAP (6.1 mg, 0.1 equiv, 0.05 mmol), and DPDTC (130. mg, 1.05 equiv, 0.525 mmol). 2-MeTHF (1.0 mL, 0.5 M) was added and the vial was capped and sealed with Teflon tape. The reaction vial was placed into an aluminum heating block which was pre-heated to 95°C and the reaction was stirred vigorously for 30 min. The reaction was taken off the aluminum block, allowed to cool to rt briefly. Once cooled, the cap was removed and amine (1.05 equiv, 0.525 mmol) and H_2O (1.0 mL, 0.25 M) were quickly added. The vial was capped, taped, and set to stir for another 30 min before working up.

Scheme 4. Substrate Scope: Amide Couplings under Flow Conditions

